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**Departement Of Electronics and Communication  
Engineering**

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**TOPIC – “ACCIDENT ALERT SYSTEM”**

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# **ACCIDENT ALERT SYSTEM**

## **ACKNOWLEDGEMENT:-**

I would like to thank my project guide, mentors, and colleagues for their support and guidance for my research and development work of the Accident Detection System. I would also appreciate the support of the internet community for the resources and tutorials I utilized for the hardware interfacing and automation part.

## **ABSTRACT:-**

Accident Detection System is an embedded system that is intended for detecting accidents in a car through the use of the 3-axis accelerometer ADXL345 that is connected to the ESP32 microcontroller. Once the accident or acceleration and deceleration is detected, this system is responsible for automatic acquisition of the GPS location and automatic sending out of the email through the Make.com flow as soon as an accident occurs. The aim of this proposed system is to provide instant intervention and avoid receiving notifications about death.

## **INTRODUCTION:-**

Road accidents can be a major worldwide concern. Many victims do not get immediate assistance because there is no alert system that automatically alerts them when an accident takes place. This project requires low-cost hardware and cloud automation that automatically sends alerts the moment a collision takes place. With ESP32, connectivity to the network will not be a problem, ADXL345 will provide accurate motion detection, and GPS elements will provide accurate positioning. Make.com will send emails containing the positioning.

## **Problem Statements:-**

Delayed Response: - Many accident victims lose their lives due to delayed emergency help, especially in remote locations where manual reporting is unreliable.

Manual Failures: - When victims are unconscious or unable to communicate, traditional reporting methods fail completely, leaving them without immediate assistance.

Need for Automation: - A fully automated system is essential one that detects accidents and alerts responders instantly without human intervention. Accident is reported late. This system will counter such issues.

## **SOLUTION:-**

Create an embedded device that:-

- Reads acceleration values from ADXL345.
- Retrieves GPS longitude and latitude coordinates.
- This sends a trigger with location information to Make.com.
- Make.com sends an email with details of the accident.

## **SCOPE:-**

This system can be applied in:-

- Personal transport (cars, bikes).
- Fleet management.
- Elderly/special needs transport services.
- Automatic Reporting by Insurance Systems.
- Integration with emergency services.

## TOOLS and RESOURCES:-

### Hardware:-

Sensor Detection: - Accelerometer and gyroscope sensors detect sudden impact or abnormal vehicle tilt, triggering the system.

Data Processing: - GPS module fetches precise real-time coordinates of the accident location.

### IMPACT SENSOR (ADXL 345):-

- Accelerometer (measures linear acceleration)
- Gyroscope (measures angular velocity)

### GPS or Location (NEO-6M):-

- The NEO-6M receives signals from at least GPS satellites orbiting Earth.
- The microcontroller extracts useful information (like coordinates) and can use it for tracking or sending alerts

### ESP32:-

- ESP32 is a low-cost, low-power microcontroller with integrated Wi-Fi and Bluetooth capabilities
- It has a dual-core 32-bit processor. This makes it suitable for tasks such as multitasking.
- It supports built-in Wi-Fi (802.11b/g/n), which provides connectivity to the Internet and the cloud.



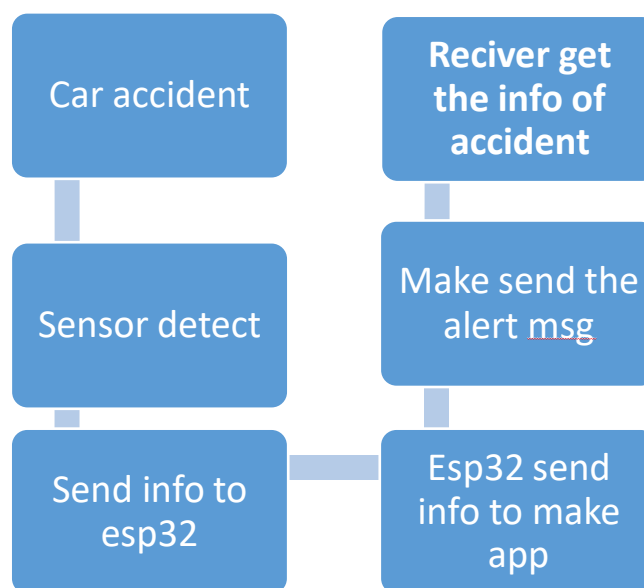
## ONLINE TOOLS:-

- Google Maps for validating coordinates
- Make.com scenario builder.

**IFTTT:** - Suited for simple, linear automation; rigidly based on a "Trigger causes Action" model, such as If I leave home, then turn off the lights.

**Make.com:-** Built for complex, multi-step workflows. Provides a visual editor called "bubble" with no step limits, branching logic via "routers", loops via "iterators", and data filtering. You'll be able to build such scenarios as: Upon form submission, check if the value is > \$100; yes - send an email, no - log to spreadsheet.

## SYSTEM ARCHITECTURE:-



**Code:-**

```
#include <WiFi.h>

#include <HTTPClient.h>

#include <Wire.h>

#include <Adafruit_Sensor.h>

#include <Adafruit_ADXL345_U.h>

#include <TinyGPS++.h>

// 1. Wi-Fi Credentials
const char* ssid = "iQOO Z9 5G";
const char* password = "mounish5789";

// 2. Make.com Webhook URL
String make_webhook_url =
"https://hook.eu1.make.com/npg3ii363slduahhttpo3yhkmg03644r3";

// 3. Settings
const float CRASH_THRESHOLD_G = 4.0; // G-Force threshold
const int TEST_BUTTON_PIN = 4;    // Pin D4 connected to button ->
Adafruit_ADXL345_Unified accel = Adafruit_ADXL345_Unified(12345);
TinyGPSPlus gps;

HardwareSerial gpsSerial(2); // Use UART2 for GPS
bool alertSent = false;

void connectToWiFi();

void sendWebhook(String lat, String lon);
```

```
void setup() {  
  Serial.begin(115200);  
  Serial.println("\n>>> ESP32 Accident Detection System Booting...");  
  gpsSerial.begin(9600, SERIAL_8N1, 16, 17);  
  Serial.println("GPS Serial Started on RX:16, TX:17");  
  pinMode(TEST_BUTTON_PIN, INPUT_PULLUP);  
  if (!accel.begin()) {  
    Serial.println("ERROR: ADXL345 not found! Check wiring.");  
    while (1) { delay(10); }  
  }  
  Serial.println("ADXL345 Found & Ready.");  
  accel.setRange(ADXL345_RANGE_16_G);  
  connectToWiFi();  
}  
  
void loop() {  
  while (gpsSerial.available() > 0) {  
    gps.encode(gpsSerial.read());  
  }  
  
  sensors_event_t event;  
  accel.getEvent(&event);  
  
  // Convert m/s^2 to G's (divide by 9.8)
```

```

float x_g = event.acceleration.x / 9.8;
float y_g = event.acceleration.y / 9.8;
float z_g = event.acceleration.z / 9.8;
// Calculate total G-force vector
float total_g = sqrt(sq(x_g) + sq(y_g) + sq(z_g));
bool crashDetected = (total_g > CRASH_THRESHOLD_G);
bool buttonPressed = (digitalRead(TEST_BUTTON_PIN) == LOW);
if ((crashDetected || buttonPressed) && !alertSent) {
    Serial.println("\n!!! TRIGGER DETECTED !!!");
    if(crashDetected) Serial.println("Cause: High G-Force Impact");
    if(buttonPressed) Serial.println("Cause: Manual Test Button");
    alertSent = true;
    String latStr, lonStr;
    if (gps.location.isValid()) {
        latStr = String(gps.location.lat(), 6);
        lonStr = String(gps.location.lng(), 6);
        Serial.print("GPS Lock Found: ");
    } else {
        // Use 0,0 if GPS has no lock yet
        latStr = "0";
        lonStr = "0";
        Serial.print("No GPS Lock (Using Dummy 0,0): ");
    }
}

```

```

}
    Serial.print(latStr); Serial.print(", "); Serial.println(lonStr);
    sendWebhook(latStr, lonStr);
    Serial.println("System locked for 15 seconds to prevent
spamming...");
    delay(15000);
    alertSent = false; // Re-arm system
    Serial.println("System Re-Armed and Ready.\n");
}
delay(50);
}

void connectToWiFi() {
    Serial.print("Connecting to Wi-Fi: ");
    Serial.println(ssid);
    WiFi.begin(ssid, password);
    int retries = 0;
    while (WiFi.status() != WL_CONNECTED) {
        delay(500);
        Serial.print(".");
        retries++;
        if(retries > 20) {
            Serial.println("\nWi-Fi Connection Failed. Will try again in loop.");

```

```
        break;
    }
}

if(WiFi.status() == WL_CONNECTED){
    Serial.println("\nWi-Fi Connected!");
}
}

void sendWebhook(String lat, String lon) {
    if(WiFi.status() == WL_CONNECTED) {
        HTTPClient http;

        String url = make_webhook_url + "?lat=" + lat + "&lon=" + lon;
        Serial.println("Sending Webhook...");
        http.begin(url);

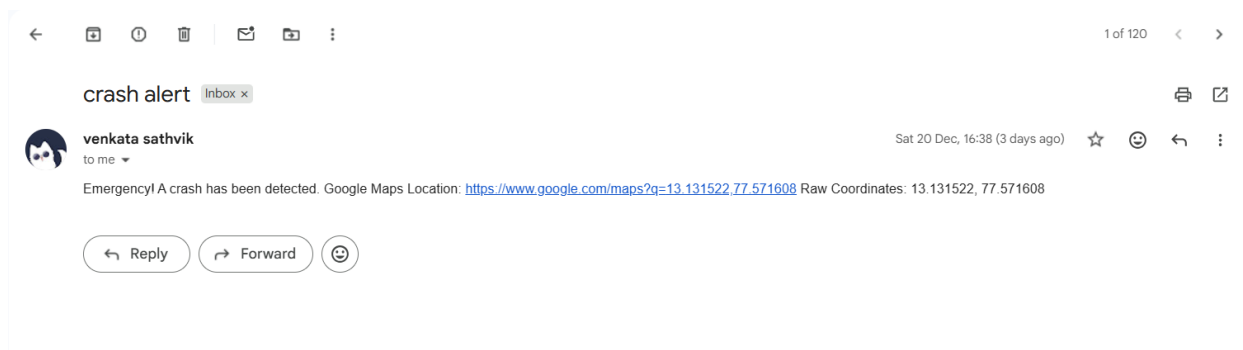
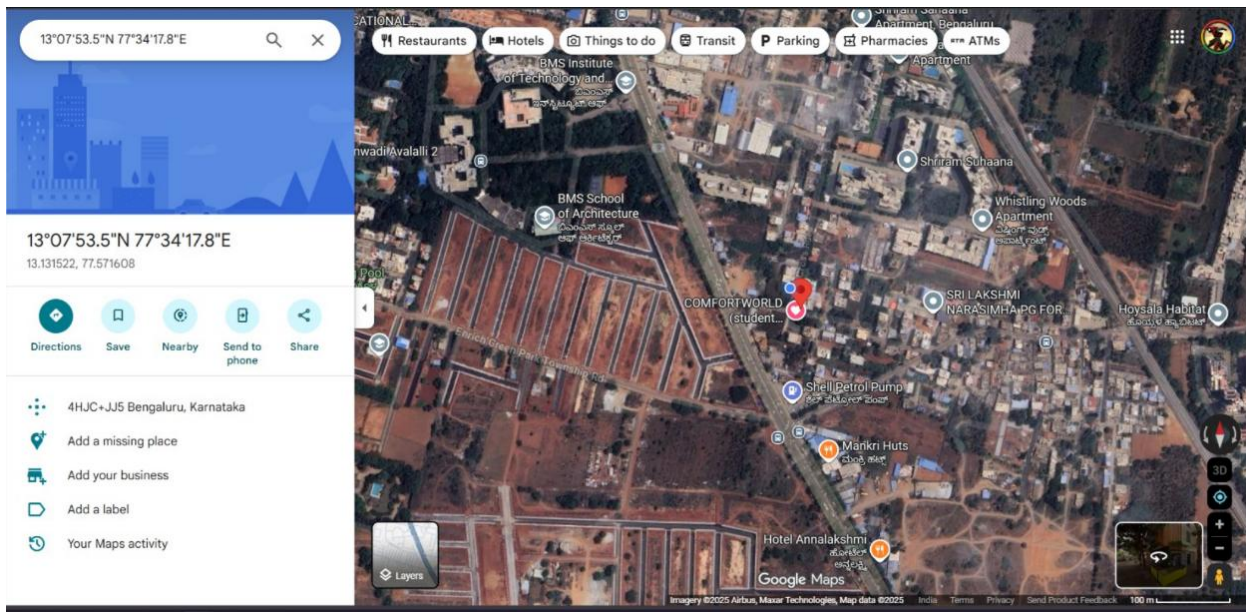
        int httpResponseCode = http.GET();
        if (httpResponseCode > 0) {
            Serial.print("HTTP Response code: ");
            Serial.println(httpResponseCode);
        } else {
            Serial.print("Error code: ");
            Serial.println(httpResponseCode);
        }
        http.end();
    }
}
```

```

    } else {
        Serial.println("WiFi Disconnected. Cannot send webhook.");
    }
}

```

## Output :-



Raw Coordinates: 13.131522, 77.571608

## **IMPLICATIONS:-**

- Optimizes Response Times for Emergencies
- Might combat deaths from late reporting.
- Enables accurate locations, accuracy for aids.
- Data available for analysis - insurance/telemetry.

## **Social Impact and Outcomes:-**

- Lives Saved: Quick alerts translate to quick rescue.
- Inclusion: Benefits vulnerable road users: elderly, night drivers.
- Lowered Anxiety: There is awareness that help will be automatically notified.
- Accuracy of GPS signals can differ when traveling through tunnels.

## **CHALLENGES FACED:-**

- False positives because of road bumps (depends on the G force set).
- GPS signal acquisition time (Cold start delay).
- Network Connectivity Inconsistencies for Rural Areas:
- Reliability of power in the vehicle environment.
- Data Security: Webhooks / Email Trigger needs to be protected using encrypted transport.

## **FUTURE ENHANCEMENT:-**

- SMS / WhatsApp Notifications.
- Direct Call or Emergency Services (API integration).
- Cloud-based accident information storage in dashboards.
- Include gyroscope for better motion detection functionality.
- Companion for real-time tracking with the help of a mobile application
- Connecting GSM module to find the nearby hospitals
- Using AI voice assistant to know whether the user is safe after the accident.

## **CONCLUSION:-**

Accident Detection System uses both cost-efficient hardware components and cloud-based automation methods that make the system efficient for implementing an emergency alerting system. Based on accurate sensor information and email automation via Make.com, the system increases security and response in case of an auto accident.

## **REFERENCES:-**

### **GitHub**

General GitHub homepage (for code hosting, repos, docs, etc.):

↪ <https://github.com> — GitHub main site

GitHub Webhooks documentation (GitHub's own webhook docs):

↪ <https://docs.github.com/en/webhooks>

### **Make.com API & Webhooks Documentation**

↪ <https://developers.make.com/api-documentation>

### **Wi-Fi and HTTPClient Tutorials by Arduino**

↪ <https://www.arduino.cc/en/Tutorial/WiFiWebClient> Arduino